Form S-6 Interactive Grid and Wind Fields

Sensitivity and Uncertainty Analysis over Southern Florida

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1 Purpose

This project implements the grid and wind-field machinery underlying *Form S-6 (Hypothetical Events for Sensitivity and Uncertainty Analysis)* of the Florida Commission on Hurricane Loss Projection Methodology Report of Activities (ROA), which supports *Standard S-3 (Uncertainty Analysis for Hurricane Model Output)*. The deliverable is an interactive browser application (HTML/CSS/JavaScript + Leaflet) that draws the official 21×40 grid over southern Florida, runs selectable hurricane wind-field models over it, and performs the sensitivity (SRC) and uncertainty (EPR) analyses the ROA prescribes.

2 The grid and storm track

The grid (ROA Figs. 3–4) has $21 \times 40 = 840$ vertices at ∼3 statute-mile spacing: east–west $E = 0, 3, \ldots, 117$ (40 columns) and north–south $N = -15, -12, \ldots, 45$ (21 rows). Of the 840 vertices, **682 are land** and 158 water, taken directly from the `Land-Water ID` worksheet of `FormS6Input.xlsx`. The point $(0, 0)$ is the storm centre at $t = 0$, nine miles east of the landfall location (25.8611°N, 80.1196°W); the storm tracks due west, $(0, 0) \to (117, 0)$, over 12 hours. Figure 1 shows the rendered grid (land vs. water), the track, and the landfall marker.

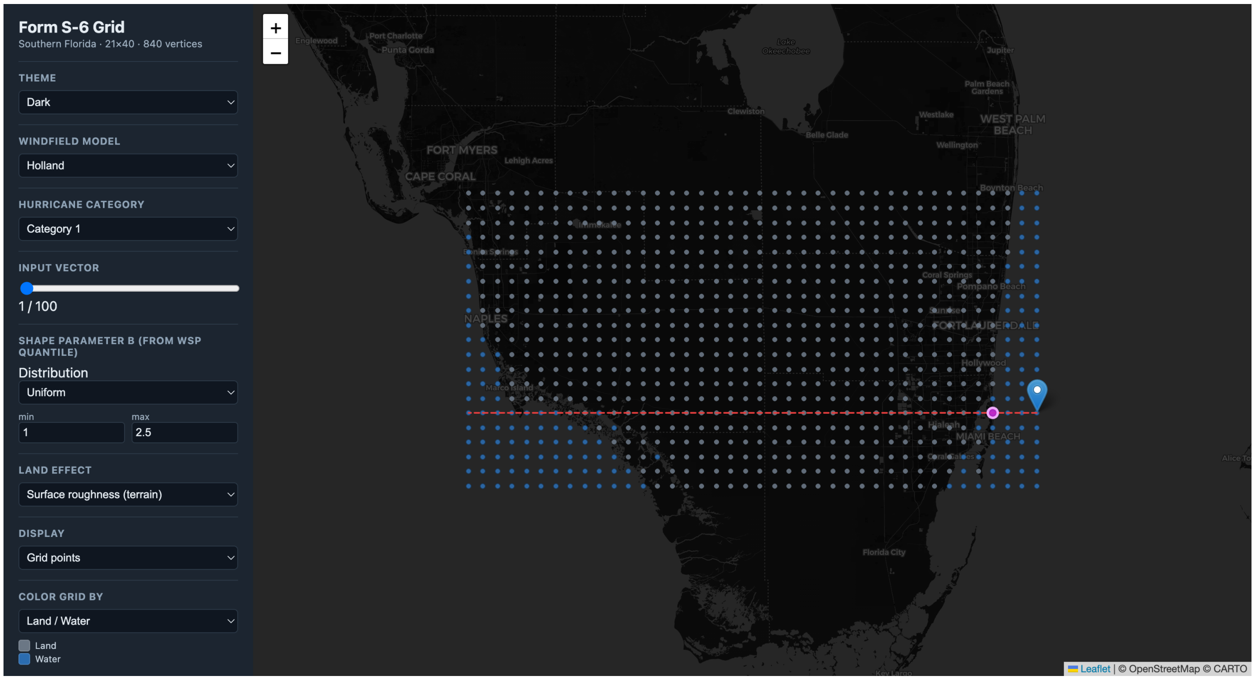

Figure 1: The 840-vertex Form S-6 grid over southern Florida: land (grey) and water (blue) vertices, the due-west storm track (dashed), the $t = 0$ centre, and the landfall marker.

3 Wind-field models

Three wind-field models are provided, each evaluated at all 840 vertices. **Powell** is an iterative boundary-layer PDE (slab) solver; **Holland** is the closed-form gradient-wind profile; **Willoughby** is a piecewise power-law profile anchored to the Holland $V_{\max}$. All three include translation asymmetry from the storm motion. Powell is precomputed in Python (PyTorch/MPS, ~2.3 s per solve); Holland and Willoughby are closed-form and computed live in the browser.

Per-vector inputs come from the `SA all Variables` worksheet: central pressure CP, radius of maximum wind Rmax, translation speed VT, wind-field shape parameter WSP, conversion factor CF, and far-field pressure FFP, for 100 vectors in each of categories 1, 3, and 5. The pressure deficit is $\Delta p = \text{FFP} - \text{CP}$. WSP is supplied as a quantile $p \in [0, 1]$ and mapped to the Holland shape parameter B through a user-selectable distribution (default Uniform$[1.0, 2.5]$, $B = B_{\min} + p\,(B_{\max} - B_{\min})$).

Conversion factor (CF). The ROA CF variable converts modelled gradient winds to surface winds using a three-zone radial rule (ROA pp. 184–185): for $r < R_{\max}$, $\text{CF}_{\text{eff}} = \text{CF} \cdot r/R_{\max}$; for $R_{\max} < r < 3R_{\max}$, $\text{CF}_{\text{eff}} = \text{CF} - 0.1\,(r - R_{\max})/(2R_{\max})$; and for $r > 3R_{\max}$, $\text{CF}_{\text{eff}} = \text{CF} - 0.1$. The models are run at gradient level ($\beta_{10} = 1$) and CF performs the gradient→surface conversion exactly as specified.

Peak-wind metric and time sampling. For each vector we record the per-vertex **12-hour peak surface wind**. The storm position is sampled finely ($\Delta t = 0.1$ h); coarse hourly sampling aliases the fast westward motion and produces a spurious "comb" artifact in the peak field, which fine sampling removes. Figures 2 and 3 show the peak field as coloured points and as filled contours.

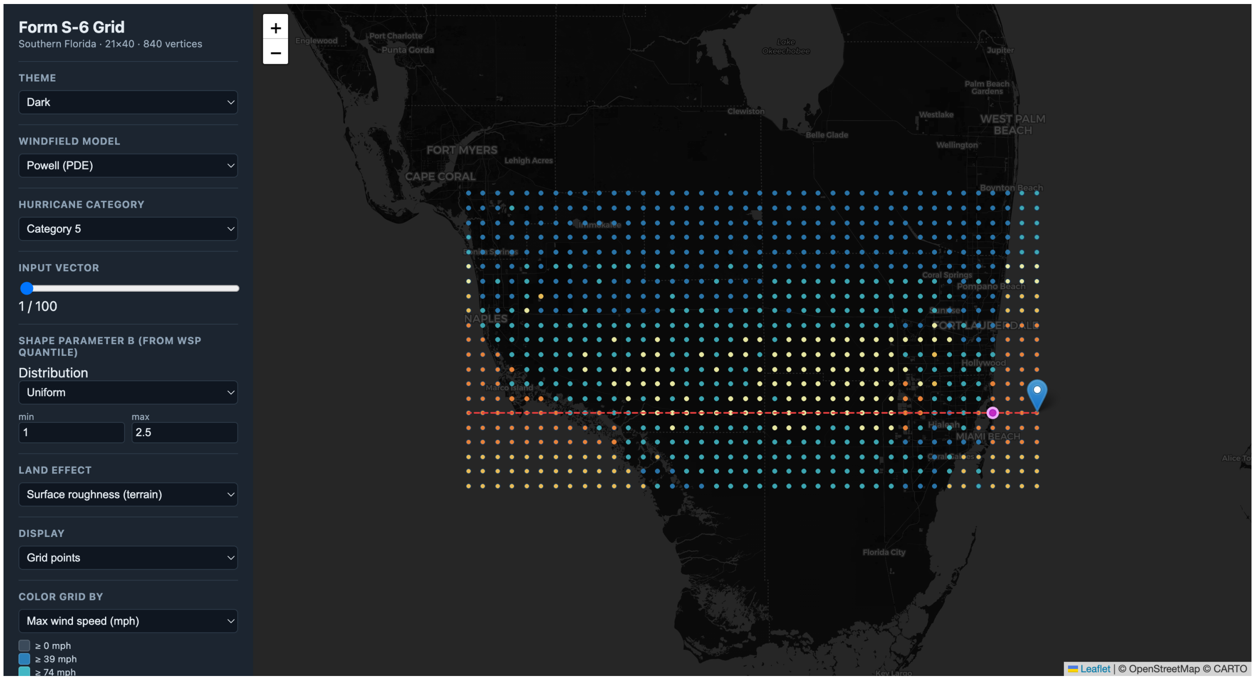

Figure 2: Powell peak surface wind (Category 5, vector 1) coloured per vertex, with surface roughness applied. Strongest winds lie north of the track—correct for a westward-moving Northern-Hemisphere cyclone, where translation adds to rotation on the north side.

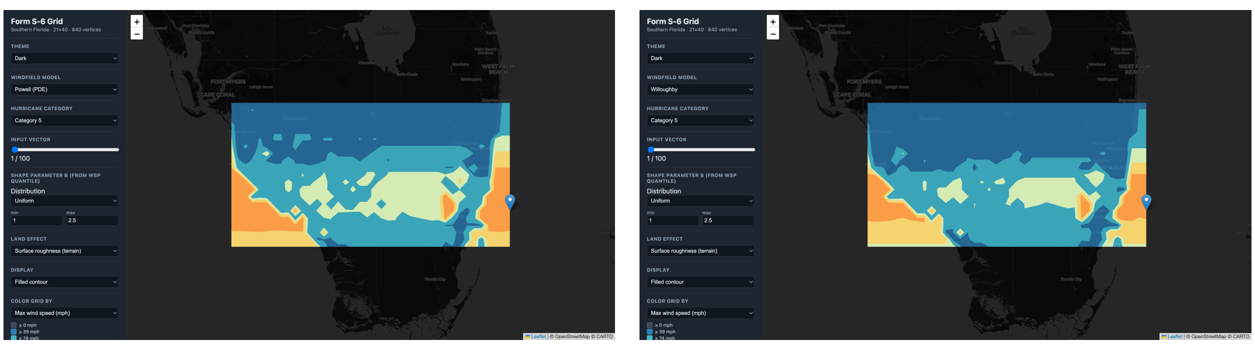

Figure 3: Filled-contour display (ROA Figs. 6–8 style) of the Category 5 peak field: Powell (left) and Willoughby (right). Contours are built on the 21×40 lattice with the wind-speed colour bands.

4 Land effect: roughness and decay

A single *land-effect* selector offers three mutually exclusive treatments of the storm's interaction with the land: **None** (marine winds), **Surface roughness**, and **Kaplan–DeMaria decay**.

Surface roughness. Roughness is applied as a per-vertex, wind-speed-independent multiplier on the final surface wind. The roughness length z_0 at each vertex is sampled from **NLCD 2021 land cover** (fetched as a properly georeferenced GeoTIFF from the MRLC WCS) via a published land-cover→ z_0 table, then converted from marine to terrain exposure with the gradient-tied **log-law**

(Vickery et al. 2009 / ESDU):

$$\text{factor} = \frac{\ln(z_{\text{ref}}/z_{0,\text{land}})/\ln(z_g/z_{0,\text{land}})}{\ln(z_{\text{ref}}/z_{0,\text{marine}})/\ln(z_g/z_{0,\text{marine}})},$$

with $z_{\text{ref}} = 10\,\text{m}$ and gradient height $z_g = 500\,\text{m}$; water returns ≈ 1. The land-mean factor is ≈ 0.67 (urban ~ 0.50, wetland ~ 0.70). This is a *spatial* terrain-friction effect.

Kaplan–DeMaria decay. Roughness alone cannot reproduce a storm weakening as it moves inland, which is a *temporal* filling process. The Kaplan–DeMaria (1995) model is offered as a separate option: a one-time coastal reduction ($R = 0.90$) at first landfall, then $V(t) = V_b + (R\,V_0 - V_b)\,e^{-\alpha t}$ with $\alpha = 0.095\,\text{h}^{-1}$ and background $V_b = 30.7\,\text{mph}$ over land, with a gentle recovery ($\alpha_{\text{rec}} = 0.05\,\text{h}^{-1}$) once the centre re-emerges over the Gulf. Landfall ($\approx$E-W 9) and Gulf exit ($\approx$E-W 99) are read from the N-$S = 0$ track land mask. Figure 6 (left) shows the resulting inland decay for a Category 5 storm. The two options are kept exclusive because α already lumps surface friction with loss of the ocean energy source.

5 Sensitivity and uncertainty analysis

Following Iman, Johnson, and Schroeder (2001), as the ROA cites:

- **Sensitivity (SRC).** Standardized regression coefficients from regressing the standardized output on the six standardized inputs over the 100 SA vectors, per category. Sign gives direction, magnitude gives influence (ROA Fig. 9).
- **Uncertainty (EPR).** Expected percentage reduction in output variance per input (ROA Fig. 10). For *Powell* the EPR is the variance share

$$\text{EPR}_i = \frac{\text{Var}(Y \mid \text{only } X_i \text{ varies})}{\text{Var}(Y_{\text{SA}})} \times 100\%,$$

  computed faithfully from the six one-variable ROA "UA for X" worksheets (1,800 PDE solves); the analytic models use the $\text{EPR}_i = \text{SRC}_i^2$ approximation.

The output metric is the **mean peak wind over the 682 land vertices** (loss-cost is the additional metric in Section 6). The computed SRCs and faithful EPRs reproduce the ROA's findings: WSP dominates for Category 1 while Rmax grows to dominate for Category 5 (faithful EPR: Cat 1 WSP $\approx 54\%$, Cat 5 Rmax $\approx 57\%$), with CP negative and CF, FFP, Rmax positive (Figure 4).

Figure 4: Sensitivity (SRC, left) and uncertainty (EPR, right) for the Powell model, plotted by hurricane category—the interactive analogues of ROA Figs. 9–10. The windows are independently movable and resizable.

6 Vulnerability and loss cost

The ROA (p. 186) fixes a single insured structure at *every* one of the 682 land vertices: a $100,000, zero-deductible, single-family, single-story, masonry, gable-roof, no-shutter home with truss anchors, shingles over 15# felt, a $\frac{1}{2}$ in. plywood deck (8d nails at 6 in. edge / 12 in. field), built 1995, roof age unknown. Because the structure is uniform, a **single vulnerability curve** applies at all points and only the wind varies.

The curve is generated from our existing HAZUS/ARA wind-vulnerability tool (`wind_vulnerability_tool_v5.html`) driven headlessly to the structure above, yielding Mean Damage Ratio (MDR) versus the ASCE 7-98 3-second peak gust at 10 m (Exposure C). The resulting masonry curve (Figure 5) is a smooth sigmoid: ∼0 below 50 mph, 50% near 100 mph, and a plateau at MDR ≈ 0.60 above ∼140 mph—the masonry ceiling (walls survive; roof, openings, and contents bound the loss), in contrast to frame construction which approaches 1.0.

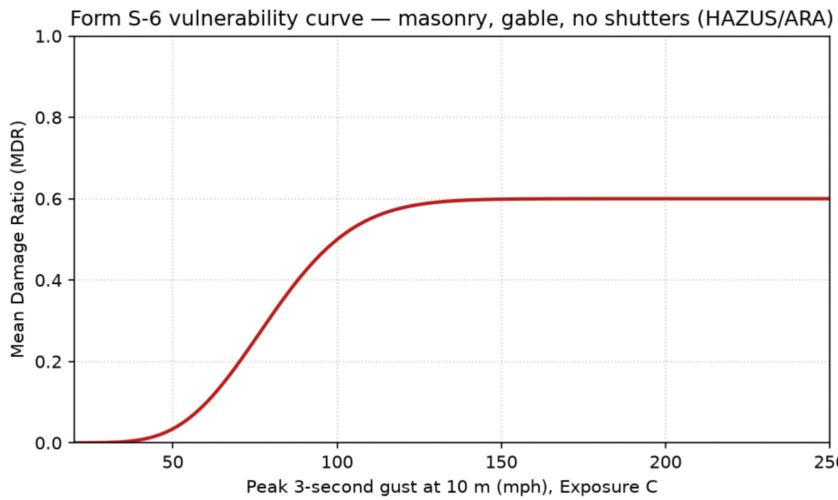

Figure 5: Form S-6 vulnerability curve (Mean Damage Ratio vs. peak 3-second gust) for the masonry, gable, no-shutter structure, generated from the HAZUS/ARA tool.

Per-vertex loss is loss = MDR(peak wind) $\times$ \$100,000; the expected loss as a percent of exposure is the total over the 682 land vertices divided by \$68,200,000 (= $682 \times$ \$100,000). The viewer colours the grid by MDR, reports total loss (\$M) and % exposure, and the hover popup shows loss % and dollars beside the peak wind. Figure 6 shows Category 5 Kaplan–DeMaria-decayed wind (left) and the corresponding loss field (right).

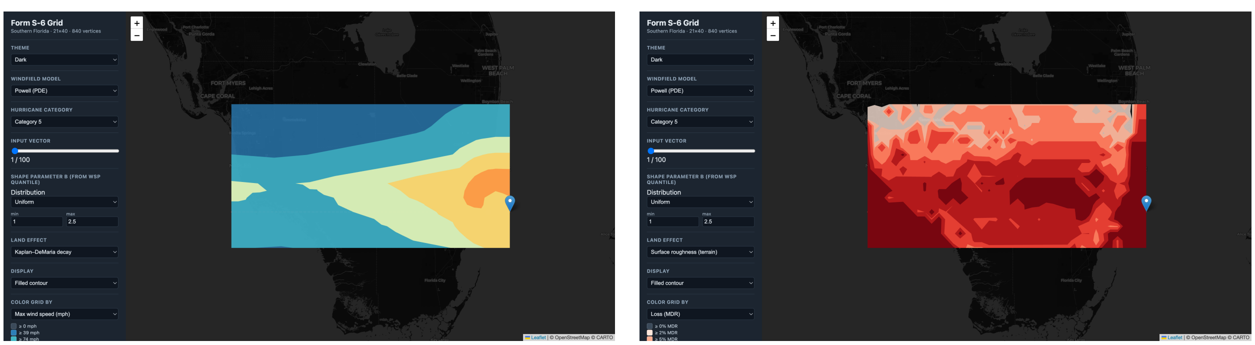

Figure 6: Category 5, Powell: Kaplan–DeMaria-decayed peak wind (left) and the loss field as Mean Damage Ratio (right), both as filled contours.

7 Interface and reproducibility

The viewer offers model / category / input-vector selection (the input vector chooses one of the 100 randomized storms for the category), the WSP$\rightarrow B$ distribution control, the three-way land-effect selector, a colour-by control (peak wind, loss MDR, or land/water), a points/filled-contour display toggle, light and dark themes (Figure 7), and nearest-point hover (peak wind, loss, coordinates, land/water, place name, and the active input parameters). **Left-clicking** a vertex opens a floating windfield popup (Figure 7) with the storm-relative isotach field—the marked vertex at its peak time—and that vertex's wind time series over the 12-hour passage. The Sensitivity and Uncertainty results open as independent, movable, resizable windows.

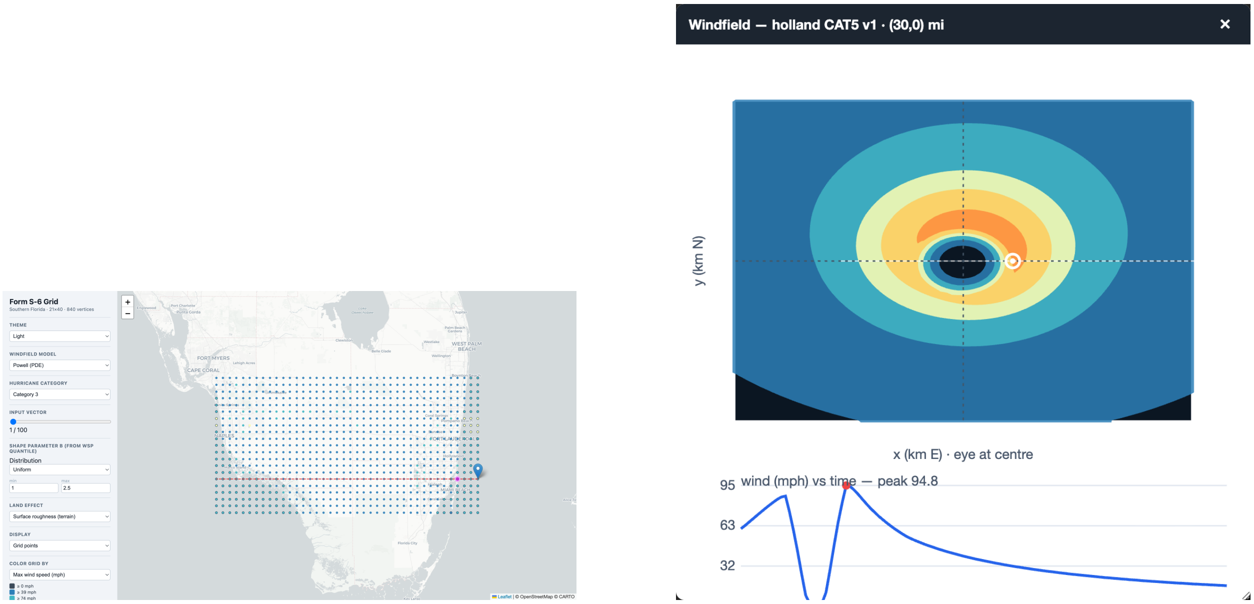

Figure 7: Left: light theme, Powell Category 3 peak-wind points. Right: the left-click windfield popup—storm-relative isotachs with the clicked vertex marked at its peak time, plus the per-vertex wind time series.

All data artifacts are reproducible via `pipeline/build_all.sh` (grid, place names, NLCD fetch, roughness, inputs, and the Powell marine/K&D/field precomputes), with `build_vulnerability.py` for the loss curve and `windfield_ua.py` for the faithful Powell EPR. The app is served with `./start` at `http://localhost:8012/web/index.html`. Figures here are captured by `docs/capture_figures.py` (Selenium driving headless Chrome against the running viewer).

References

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